

Iterated angular–radial limits settle the literal Bessel–Fourier convergence conjecture

Candidate full affirmative answer to Conjecture 1 in arXiv:2202.06119

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Status: candidate full affirmative result under the printed quantifiers; semantic review important. For every $4/3 < p < 4$ and every $f \in L^p(\mathbb{D})$, there are cofinal rectangular cutoffs (N_k, M_k) for which the Bessel–Fourier partial sums converge to f in $L^p(\mathbb{D})$. In fact, the angular cutoffs $M_k \rightarrow \infty$ may be fixed in advance and the radial cutoffs may be required to exceed any prescribed lower bounds. The proof does *not* produce one path that works for all f simultaneously; that stronger quantifier order is not stated in the conjecture and remains open.

1 Source conjecture and quantifier scope

Let $\mathbb{D} = [0, 1] \times \mathbb{T}$ in polar coordinates with measure $r \, dr \, d\theta$. Acosta Babb [1] writes the Bessel–Fourier rectangular sums as

$$S_{N,M}f(r, \theta) = \sum_{m=-M}^M \sum_{n=1}^N a_{m,n}(f) J_{|m|}(j_n^{(m)} r) e^{2\pi i m \theta}. \quad (1)$$

Conjecture 1, on arXiv PDF page 6, asks whether, for every $4/3 < p < 4$ and $f \in L^p(\mathbb{D})$,

$$\|S_{N_k, M_k}f - f\|_{L^p(\mathbb{D})} \longrightarrow 0$$

“for some appropriate choice” of $N_k, M_k \in \mathbb{N}$.

Table 1: L^p convergence of Bessel–Fourier series for various ranges of p .

In light of the existing results (Balodis and Córdoba, 1999; Wing, 1950; Benedek and Panzone, 1972/73), we might offer the following conjecture.

Conjecture 1. For all $4/3 < p < 4$ and $f \in L^p(\mathbb{D})$,

$$\lim_{k \rightarrow \infty} \|S_{N_k, M_k}f - f\|_{L^p(\mathbb{D})} = 0,$$

for some appropriate choice of $N_k, M_k \in \mathbb{N}$, $k \geq 1$. For p outside of this range, convergence fails in general.

Figure 1: Conjecture 1 in arXiv:2202.06119, PDF page 6. The image contains the complete printed statement.

The order of quantifiers printed in the article (and repeated unchanged in the author’s 2024 thesis) is

$$\forall f \in L^p(\mathbb{D}) \quad \exists (N_k, M_k)_{k \geq 1}. \quad (2)$$

The discussion immediately after the conjecture emphasizes the difficulty of uniform operator bounds, which may indicate that the stronger order

$$\exists(N_k, M_k)_{k \geq 1} \quad \forall f \in L^p(\mathbb{D}) \quad (3)$$

was informally intended. The theorem below proves (2), with two useful strengthenings, but deliberately does not claim (3).

2 Main theorem

Let $A > 0$ denote the absolute constant appearing in the Balodis–Córdoba cutoff geometry quoted in the source. It is included below to show that the diagonal path can also obey $N_k \geq AM_k + 1$; the proof itself does not need this relation.

Theorem 1 (Literal conjecture, with arbitrary lower bounds). *Let $4/3 < p < 4$, let $f \in L^p(\mathbb{D})$, and let (M_k) be any sequence of positive integers with $M_k \rightarrow \infty$. Given any positive integers L_k , there are strictly increasing integers N_k such that*

$$N_k \geq \max\{L_k, AM_k + 1\} \quad (4)$$

and

$$\|S_{N_k, M_k} f - f\|_{L^p(\mathbb{D})} \rightarrow 0. \quad (5)$$

In particular, the affirmative assertion of Conjecture 1 holds exactly as printed.

Theorem 2 (Uniformity on compact sets). *Under the hypotheses of Theorem 1, if $K \subset L^p(\mathbb{D})$ is norm compact, the cutoffs N_k can be chosen so that*

$$\sup_{f \in K} \|S_{N_k, M_k} f - f\|_{L^p(\mathbb{D})} \rightarrow 0$$

while still satisfying (4) and strict increase.

Corollary 3 (One path for any countable family). *For every countable set $\{f_1, f_2, \dots\} \subset L^p(\mathbb{D})$ and every $M_k \rightarrow \infty$, one can choose a single cofinal sequence (N_k, M_k) along which $S_{N_k, M_k} f_j \rightarrow f_j$ for every j .*

3 Proof intuition

A rectangle has two independent directions. Freeze the angular width first. Angular Fourier partial sums converge in L^p , so retaining the modes $|m| \leq M$ approximates f when M is large. For a fixed M , only finitely many radial functions f_m remain. The classical one-dimensional Fourier–Bessel theorem says that, for each fixed order $|m|$, its radial partial sums converge in exactly the range $4/3 < p < 4$. One common radial cutoff works for this finite list. Choosing that cutoff after the angular width is chosen gives an iterated limit; a diagonal selection turns it into the requested sequence of rectangles.

This mechanism needs no operator bound uniform in both indices. That is why it proves the function-dependent quantifiers (2) while leaving the whole-space uniform-path formulation (3) untouched.

4 The two one-parameter limits

For $f \in L^p(\mathbb{D})$, define its angular coefficients

$$f_m(r) = \int_{\mathbb{T}} f(r, \theta) e^{-2\pi i m \theta} d\theta \quad (m \in \mathbb{Z}).$$

Hölder's inequality on the probability space $\mathbb{T}$ gives

$$\|f_m\|_{L^p((0,1),r dr)} \leq \|f\|_{L^p(\mathbb{D})}. \quad (6)$$

Let

$$P_M f(r, \theta) = \sum_{|m| \leq M} f_m(r) e^{2\pi i m \theta}. \quad (7)$$

Lemma 4 (Angular limit). *For $1 < p < \infty$,*

$$\|P_M f - f\|_{L^p(\mathbb{D})} \longrightarrow 0.$$

Moreover, $\sup_M \|P_M\|_{L^p(\mathbb{D}) \rightarrow L^p(\mathbb{D})} < \infty$.

Proof. The M. Riesz theorem gives the uniform $L^p(\mathbb{T})$ bound for ordinary Fourier partial sums. Apply it to $f(r, \cdot)$ and integrate the p th power against $r dr$. Convergence follows first for finite angular trigonometric polynomials with radial L^p coefficients and then for all f by density and the uniform bound. $\square$

For a nonnegative integer ν , write B_N^ν for the first N terms of the one-dimensional Fourier–Bessel expansion of order ν on $L^p((0,1), r dr)$. The classical mean-convergence theorem of Wing and Benedek–Panzone [2, 3] says that, for each fixed ν ,

$$B_N^\nu g \longrightarrow g \quad \text{in } L^p((0,1), r dr), \quad \frac{4}{3} < p < 4. \quad (8)$$

This is precisely the fixed-order result invoked in the source article.

Lemma 5 (Fixed angular window). *If $4/3 < p < 4$ and M is fixed, then*

$$\|S_{N,M} f - P_M f\|_{L^p(\mathbb{D})} \longrightarrow 0 \quad (N \rightarrow \infty).$$

For this fixed M , the operators $S_{N,M} - P_M$ are uniformly bounded in N on $L^p(\mathbb{D})$.

Proof. The definitions give the exact separation

$$S_{N,M} f - P_M f = \sum_{|m| \leq M} (B_N^{|m|} f_m - f_m) e^{2\pi i m \theta}. \quad (9)$$

Since multiplication by $e^{2\pi i m \theta}$ embeds $L^p((0,1), r dr)$ isometrically into $L^p(\mathbb{D})$, the triangle inequality and (8) give

$$\|S_{N,M} f - P_M f\|_{L^p(\mathbb{D})} \leq \sum_{|m| \leq M} \|B_N^{|m|} f_m - f_m\|_{L^p((0,1), r dr)} \longrightarrow 0. \quad (10)$$

For every fixed order, pointwise strong convergence in (8) and Banach–Steinhaus imply

$$\sup_N \|B_N^{|m|}\| < \infty.$$

Combine these finitely many bounds with (6) and (9). $\square$

5 Proofs of the main results

Proof of Theorem 1. For every k , Lemma 5 permits us to choose N_k so large that it is strictly larger than N_{k-1} , satisfies (4), and

$$\|S_{N_k, M_k} f - P_{M_k} f\|_{L^p(\mathbb{D})} < 2^{-k}. \quad (11)$$

Then

$$\|S_{N_k, M_k} f - f\|_p \leq 2^{-k} + \|P_{M_k} f - f\|_p \rightarrow 0$$

by Lemma 4. □

Proof of Theorem 2. We use twice the standard fact that uniformly bounded operators converging strongly converge uniformly on every norm-compact set. Lemma 4 therefore gives

$$\sup_{f \in K} \|P_{M_k} f - f\|_p \rightarrow 0.$$

For each fixed k , Lemma 5 and its uniform boundedness statement give

$$\sup_{f \in K} \|S_{N, M_k} f - P_{M_k} f\|_p \rightarrow 0 \quad (N \rightarrow \infty).$$

Choose N_k successively to make the latter supremum smaller than 2^{-k} and to meet the lower bounds. The triangle inequality finishes the proof. □

Proof of Corollary 3. At stage k , apply the finite-set version of Theorem 2 to $\{f_1, \dots, f_k\}$ and choose N_k so the radial error is below 2^{-k} for all these functions. For each fixed j , the angular error tends to zero as $k \rightarrow \infty$, and the radial bound applies whenever $k \geq j$. □

6 Quantifier caveat and the negative clause

The proof selects N_k after seeing f (or a compact/countable family). It therefore does not show

$$\sup_k \|S_{N_k, M_k}\|_{L^p(\mathbb{D}) \rightarrow L^p(\mathbb{D})} < \infty,$$

which by Banach–Steinhaus is the natural requirement for one fixed path to converge for every f . The source’s comment about the difficulty of uniform bounds concerns exactly this stronger issue.

The sentence “for p outside this range, convergence fails in general” is also sensitive to the same semantics. Classical mean-convergence results show failure of the full fixed-order partial-sum sequence outside the open range (including endpoint failure in the standard uniform-boundedness sense). They do not automatically exclude every f -dependent sparse subsequence. This packet proves the affirmative clause exactly as printed; it does not make a new claim about that adaptive-subsequence interpretation of the negative clause.

7 Focused proof and upgrade audits

Eight focused checks were made before promotion:

1. The arXiv source, 2023 published version, and 2024 author thesis were checked; all retain the quantifier “for every f , for some choice.”

2. The angular coefficient map was checked directly in (6), so no unjustified pointwise L^p trace is used.
3. The angular limit was reduced to the scalar M. Riesz theorem and product density, avoiding any false Plancherel step for $p \neq 2$.
4. The fixed-order radial theorem was checked against the classical Fourier–Bessel literature and the source’s own stated range.
5. The exact separation (9) was checked against the normalization and order convention $J_m = J_{|m|}$ in the source.
6. The finite-mode triangle estimate (10) avoids any constant uniform in the angular order; finiteness is enough.
7. The diagonal choice was strengthened to arbitrary prescribed M_k , arbitrary radial lower bounds, compact families, and countable families.
8. A universal whole-space path was considered. The argument stops precisely at the missing uniform bound in both indices, so that stronger problem is explicitly not claimed.

8 Novelty check, limitations, and review recommendation

A bounded search on 13 August 2026 covered the exact title and conjecture wording, “iterated limit,” “diagonal,” and “fixed angular” Bessel–Fourier searches, the 2023 publication, the author’s 2024 thesis, the citing triangular-domain paper, arXiv records through 2026, and the author’s 2026 coefficient-decay paper. The thesis still labels the problem open, and no later solution or this diagonal argument was found. Novelty is plausible, not certified, and priority is not claimed.

The mathematical proof has no computational dependency. Its limitation is semantic rather than technical: it settles the literal $\forall f \exists$ path assertion, not the stronger $\exists$ path $\forall f$ assertion that the surrounding discussion may have intended. Human review should first decide the intended quantifier order. Under the printed order, check only the fixed-window identity (9) and the applicability of fixed-order mean convergence; the rest is a finite-sum diagonal argument.

References

- [1] R. L. Acosta Babb, *Remarks on the L^p convergence of Bessel–Fourier series on the disc*, Comptes Rendus Mathématique **361** (2023), 1075–1080; arXiv:2202.06119, doi:10.5802/crmath.464.
- [2] G. M. Wing, *The mean convergence of orthogonal series*, Amer. J. Math. **72** (1950), 792–808.
- [3] A. Benedek and R. Panzone, *Mean convergence of series of Bessel functions*, Rev. Unión Mat. Argentina **26** (1972/73), 42–61.